

Structural Limits for AI Scaling

Applying Insights of Recursive Triadic Fractal Universe

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May 2026

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Abstract

Large language models exhibit remarkably smooth scaling behaviour, yet recent empirical studies show clear signs of saturation: diminishing returns in loss reduction, representation interference, instability at long context lengths, and a breakdown of compute–data optimality. Although these phenomena are well documented, the field lacks a theory-neutral architectural explanation for why monolithic transformer models encounter such limits.

This paper applies the structural framework introduced in *The Triadic Cosmos: The Universe as Recursive Triadic Fractal* to the domain of AI scaling. Within this framework, computational systems are governed by a balance between three structural roles—energy-like, geometry-like, and information-like components. We show that transformer-based LLMs systematically violate this balance: they scale energy and information dimensions aggressively while lacking the geometric structure required for stable, coherent, and interference-resistant computation. This imbalance provides a unified architectural explanation for the empirical limits observed in current scaling laws.

We contrast this with the *Dynamic Modular Language Model* (DMLM) architecture, which enforces triadic balance through paging, evaluator plurality, and modular canonical forms. DMLM thereby avoids several structural bottlenecks inherent to monolithic transformers and offers a pathway toward architectures that scale in depth, coherence, and reasoning ability.

The contribution of this paper is not a new scaling law, but a structural account of why existing laws break down and why certain architectural directions remain viable. By grounding AI scaling behaviour in a general triadic framework, we provide a conceptual basis for understanding the limits of current models and for designing next-generation systems that scale more robustly.

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1 Introduction

The rapid progress of large language models (LLMs) over the past five years has been driven primarily by the empirical regularities known as *scaling laws*. As model size, dataset size, and compute increase, performance improves in a predictable manner, enabling the construction of increasingly capable systems. However, recent frontier-scale experiments reveal that this trend is not indefinite: transformer-based LLMs exhibit clear signs of saturation, including diminishing returns in loss reduction, interference between representations, instability at long context lengths, and a breakdown of compute-data optimality. These observations raise a fundamental question: *why* do monolithic transformer architectures encounter such limits?

Existing analyses provide valuable empirical characterizations but do not offer a theory-neutral architectural explanation. Most accounts attribute the observed limitations to engineering constraints, optimization difficulties, or insufficient data. Yet the consistency of the phenomena across model families suggests that deeper structural factors are at play. Understanding these factors is essential not only for interpreting current scaling behaviour, but also for identifying viable directions for next-generation architectures.

This paper approaches the problem from a structural perspective. It applies the framework introduced in *The Triadic Cosmos: The Universe as Recursive Triadic Fractal*, which proposes that physical, informational, and computational systems share a common organization based on three interacting structural roles: energy-like, geometry-like, and information-like components. Although originally developed in the context of physical ontology, the framework is theory-neutral and applies equally to artificial systems. When viewed through this lens, transformer-based LLMs can be understood as architectures that scale energy-like and information-like dimensions aggressively while lacking the geometric structure required for stable, coherent, and interference-resistant computation.

The central claim of this paper is that the empirical limits of LLM scaling arise from this structural imbalance. We show that monolithic transformers violate triadic balance conditions, leading to representational saturation, unstable long-range interactions, and inefficient generalization dynamics. These limitations are not incidental but follow from the architectural form itself.

To illustrate the implications of this perspective, we contrast transformer scaling with the *Dynamic Modular Language Model* (DMLM) architecture, which enforces triadic balance through modularity, paging, evaluator plurality, and canonical forms. DMLM thereby avoids several structural bottlenecks inherent to monolithic models and offers a pathway toward architectures that scale more robustly in depth, coherence, and reasoning ability.

The goal of this paper is not to introduce a new empirical scaling law, but to provide a structural account of why existing laws break down and why certain architectural directions remain viable. By grounding AI scaling behaviour in a general triadic framework, we aim to clarify the conceptual landscape and support the development of next-generation systems that scale beyond the limits of current transformer-based approaches.

2 Related Work

Research on the limits of large-scale machine learning has expanded rapidly in recent years, driven by the empirical observation that transformer-based language models exhibit predictable scaling behaviour but also clear diminishing returns at frontier scales. This section reviews the most relevant strands of work and situates the present paper within that landscape.

2.1 Scaling Laws for Large Language Models

The modern study of AI scaling began with the empirical laws introduced by Kaplan et al. Kaplan et al. 2020, who demonstrated that loss decreases as a smooth power law in model size,

dataset size, and compute. Subsequent work refined these results, most notably the Chinchilla scaling laws of Hoffmann et al. Hoffmann, Borgeaud, Mensch, et al. 2022, which showed that optimal performance requires balancing model parameters and dataset size. These findings motivated the current generation of frontier LLMs.

More recent analyses have revealed structural limitations of transformer scaling. DeepMind’s investigations into phase transitions and generalization dynamics Anonymous 2024a suggest that improvements slow dramatically once models enter a high-capacity regime. Anthropic’s studies on context length, interference, and representation saturation Anonymous 2024b; Anonymous 2025 further indicate that transformers face intrinsic geometric and informational bottlenecks that are not resolved by brute-force scaling.

Together, these works provide strong empirical evidence that monolithic transformer architectures encounter fundamental constraints, but they do not offer a theory-neutral architectural explanation for why these limits arise.

2.2 Architectural Alternatives and Modular Approaches

A parallel line of research explores modular or hierarchical architectures as alternatives to monolithic transformers. Mixture-of-experts models Shazeer, Mirhoseini, Maziarz, et al. 2017; Lepikhin, Lee, Xu, et al. 2021 improve parameter efficiency but do not address deeper structural issues such as representational interference or the absence of stable geometric organization. Neurosymbolic and hybrid reasoning systems Li et al. 2023 attempt to extend LLM capabilities but remain ad hoc and lack a unifying architectural principle.

The *Dynamic Modular Language Model* (DMLM) architecture proposed by De Witte 2026a introduces a principled modular framework based on paging, evaluator plurality, and canonical forms. DMLM provides a contrasting example of a system that maintains stability and scalability by enforcing structural separation between energy-like, geometry-like, and information-like roles. This paper builds on that contrast to show why monolithic transformers violate structural balance conditions and therefore face inherent scaling limits.

2.3 Foundational Structural Frameworks

The present work is grounded in the structural ontology introduced in *The Triadic Cosmos: The Universe as a Recursive Triadic Fractal* De Witte 2026b. That foundational paper develops a theory-neutral architectural framework in which physical, informational, and computational systems share a common triadic structure (Energy–Geometry–Information) and evolve through recursive fractal organization. While the foundational paper focuses on physical unification, its structural principles apply equally to artificial systems.

The current paper extends those principles to the domain of AI scaling, showing that the empirical limitations observed in transformer-based LLMs correspond directly to violations of triadic balance and geometric coherence. In this sense, the present work is an application of the broader Triadic Cosmos framework to a concrete and technologically relevant domain.

3 Structural Limits of Monolithic LLM Scaling

Transformer-based language models have achieved unprecedented performance across a wide range of tasks, yet their behaviour at frontier scales reveals a consistent pattern of structural limitations. This section analyses these limitations through the lens of the triadic framework, showing that they arise not from optimization difficulties or insufficient data, but from architectural imbalances inherent to monolithic transformer design.

3.1 Energy-like Scaling: Compute Without Structural Constraint

Transformers scale primarily along an energy-like dimension: increasing parameter count, depth, width, and training compute. This produces predictable improvements at moderate scales, but at larger scales the absence of geometric constraint leads to unstable long-range interactions and inefficient generalization dynamics.

Attention layers aggregate information globally without an intrinsic notion of structure, locality, or hierarchical organization. As model size grows, this global coupling amplifies noise, increases sensitivity to spurious correlations, and produces interference between unrelated representations. The result is a regime in which additional compute yields diminishing returns, consistent with the observed flattening of empirical scaling curves.

3.2 Information-like Scaling: Saturation and Interference

The second axis along which transformers scale is information-like: expanding embedding dimensions, vocabulary size, and context length. However, these expansions encounter intrinsic limits. Empirical studies show that representation quality saturates as embedding spaces become crowded, leading to interference, loss of distinguishability, and context-length instability.

Because transformers lack mechanisms for structural separation or canonical organization of representations, new information is superimposed on existing representations in a high-dimensional but unstructured space. This produces characteristic failure modes:

- **representation interference**, where semantically unrelated tokens compete for overlapping subspaces;
- **context degradation**, where long sequences accumulate noise faster than signal;
- **instability under scaling**, where increasing context length or embedding size yields sublinear or negative returns.

These effects are not artifacts of training but consequences of the architectural absence of geometry-like structure.

3.3 Geometric Deficiency: The Missing Structural Role

The triadic framework identifies a third structural role: geometry-like components that enforce organization, separation, and stability. In physical systems, geometric structure constrains interactions, preserves distinguishability, and prevents uncontrolled coupling. In computational systems, analogous mechanisms include modularity, hierarchical organization, and explicit structural separation.

Monolithic transformers lack such geometric structure. Attention provides a uniform, undifferentiated interaction mechanism, and the model state evolves in a single, globally coupled representational space. Without geometric constraints, the system cannot maintain stable attractors, canonical forms, or long-range coherence. This deficiency explains several well-documented phenomena:

- **loss of coherence** in long reasoning chains;
- **instability** when scaling depth or context length;
- **failure to maintain modular or compositional structure**;
- **difficulty with planning, recursion, and multi-step reasoning**.

These limitations arise not from insufficient scale but from the absence of the geometric role required for stable computation.

3.4 Triadic Imbalance as the Source of Scaling Limits

Taken together, these observations reveal a structural imbalance: transformers scale energy-like and information-like dimensions aggressively while leaving the geometry-like dimension effectively absent. This imbalance leads to:

- **representational saturation** (information-like overload),
- **global coupling and noise amplification** (energy-like excess),
- **lack of structural coherence** (geometry-like deficiency).

The empirical limits of transformer scaling—diminishing returns, interference, context instability, and reasoning failures—follow directly from this imbalance. They are not incidental engineering challenges but structural consequences of the monolithic architecture itself.

This analysis motivates the search for architectures that restore triadic balance. In Section 4, we examine the Dynamic Modular Language Model (DMLM) as an example of such an architecture, demonstrating how modularity, paging, evaluator plurality, and canonical forms provide the geometric structure missing in transformers and thereby avoid the scaling limits identified here.

4 DMLM as a Triadic, Modular Alternative

The structural limitations identified in monolithic transformer scaling motivate the search for architectures that maintain stability and coherence as they grow. The *Dynamic Modular Language Model* (DMLM) architecture provides an example of such a system. Rather than scaling a single globally coupled model, DMLM organizes computation through modularity, paging, evaluator plurality, and canonical forms. These mechanisms collectively instantiate the geometry-like structural role that transformers lack, restoring triadic balance and enabling more robust scaling behaviour.

4.1 Modularity and Structural Separation

DMLM decomposes computation into a collection of specialized micro-models, each responsible for a distinct representational or functional domain. This modularity prevents the uncontrolled superposition of representations that characterizes monolithic transformers. By separating representational subspaces, DMLM reduces interference, preserves distinguishability, and enables stable long-range interactions.

Modularity also introduces a form of structural locality: interactions occur within and between modules according to explicit organizational constraints rather than global attention. This provides the geometric structure required for stable computation at scale.

4.2 Paging as a Geometric Mechanism

A central component of DMLM is its paging mechanism, which governs how information is routed, stored, and retrieved across modules. Paging imposes a geometric organization on the model’s internal state: information is not globally accessible but is structured into discrete, addressable regions. This prevents global coupling, reduces noise amplification, and ensures that long-range dependencies are mediated through controlled pathways.

Paging thereby plays the role of a geometry-like constraint, analogous to spatial organization in physical systems. It limits the effective interaction radius, stabilizes representational dynamics, and supports coherent multi-step computation.

4.3 Evaluator Plurality and Information-like Structure

DMLM employs multiple evaluators—distinct mechanisms that assess, refine, or select representations produced by micro-models. This evaluator plurality introduces an information-like dimension that is structurally separated from both the energy-like micro-models and the geometry-like paging system.

By distributing evaluative functions across multiple specialized components, DMLM avoids the bottlenecks and saturation effects that arise when a single monolithic model must simultaneously represent, evaluate, and integrate all information. Evaluator plurality enhances robustness, reduces interference, and supports compositional reasoning.

4.4 Canonical Forms and Stable Attractors

A key feature of DMLM is the use of canonical forms—standardized, stable representational structures that serve as attractors for the system’s dynamics. Canonical forms provide a consistent interface between modules, enabling compositionality and preventing drift in representational space.

In contrast to transformers, where representations evolve in a single high-dimensional space without stable anchors, DMLM’s canonical forms enforce structural coherence. They ensure that information remains interpretable across modules and across multiple computational steps, supporting depth, stability, and reuse.

4.5 Restoring Triadic Balance

Taken together, modularity, paging, evaluator plurality, and canonical forms restore the triadic balance absent in monolithic transformers:

- **energy-like role:** micro-models provide localized computational power;
- **geometry-like role:** paging and modular structure constrain and organize interactions;
- **information-like role:** evaluators and canonical forms maintain coherence and distinguishability.

This balanced organization avoids the structural bottlenecks identified in Section 3. DMLM does not rely on global coupling, does not accumulate interference in a single representational space, and does not require ever-increasing compute to compensate for geometric deficiency. Instead, it scales by recursive expansion of modular structure, enabling growth in capacity without loss of stability or coherence.

The purpose of this section is not to advocate for a specific implementation, but to illustrate how architectures that respect triadic balance can avoid the structural limits inherent to monolithic transformer scaling. DMLM serves as one concrete example of such an approach.

5 Testable Predictions for AI Architectures

The foundational paper *The Triadic Cosmos: The Universe as Recursive Triadic Fractal* formulates several structural predictions that follow from the cyclic interdependence of energy, geometry, and information. This section reformulates the predictions most relevant to computational systems by introducing an intermediate *AI Prediction* that translates each structural principle into a falsifiable consequence for artificial architectures. For each prediction, we contrast the expected behaviour of monolithic transformer LLMs with that of modular architectures such as the Dynamic Modular Language Model (DMLM).

5.1 Prediction 1: Universal Area–Law Scaling

Triadic Prediction. Informational capacity is bounded by geometric structure; effective capacity scales with boundary structure rather than volume.

AI Prediction. Architectures without geometric separation will experience representational saturation as dimensionality increases, while modular architectures will scale capacity through boundary expansion.

LLM Implication (Testable).

- sublinear returns from increasing embedding size,
- interference growth in dense representational regions,
- diminishing returns in loss reduction at frontier scales.

DMLM Implication (Testable).

- capacity increases by adding modules (boundary expansion),
- reduced interference through structural separation,
- fractal-like scaling of informational capacity.

5.2 Prediction 2: Emergent Time from Triadic Interdependence

Triadic Prediction. Temporal evolution arises from the interplay of energetic, geometric, and informational roles; time cannot be an independent primitive.

AI Prediction. Architectures lacking recursive structure will fail to maintain coherent multi-step computation, while architectures with explicit recursion will support stable temporal evolution.

LLM Implication (Testable).

- chain-of-thought instability,
- absence of stable intermediate states,
- degradation in multi-step reasoning as sequence length grows.

DMLM Implication (Testable).

- stable intermediate representations via PRE recursion,
- coherent multi-step reasoning,
- improved temporal structure as modules scale.

5.3 Prediction 3: Minimal Geometric Resolution

Triadic Prediction. Geometric structure cannot be arbitrarily fine; systems require minimal resolvable units to maintain distinguishability.

AI Prediction. Architectures that attempt to scale context or embedding resolution without structural units will encounter collapse, while modular architectures will maintain resolution through discrete representational units.

LLM Implication (Testable).

- instability at ultra-long context lengths,
- representational collapse in dense embeddings,
- diminishing returns from context-length scaling.

DMLM Implication (Testable).

- stable resolution via module-level units,
- context scaling through paging rather than embedding expansion,
- avoidance of collapse through discrete representational boundaries.

5.4 Prediction 4: Causal–Informational Coherence

Triadic Prediction. Informational influence must respect geometric causal structure; systems without locality exhibit uncontrolled coupling.

AI Prediction. Architectures with global coupling will amplify noise and lose compositionality, while architectures with structural locality will maintain coherent interactions.

LLM Implication (Testable).

- global noise propagation via attention,
- loss of compositional structure,
- degradation of long-range dependencies.

DMLM Implication (Testable).

- localized interactions within modules,
- preserved compositionality across modules,
- stable long-range dependencies through controlled pathways.

5.5 Prediction 5: Embedded Observers and Informational Boundaries

Triadic Prediction. Observers must be embedded informational–energetic subsystems; evaluation cannot be external to the architecture.

AI Prediction. Architectures that conflate representation and evaluation will struggle with self-monitoring, while architectures with internal evaluators will support self-correction.

LLM Implication (Testable).

- no internal evaluative subsystems,
- conflation of representation and evaluation,
- difficulty with self-monitoring or iterative refinement.

DMLM Implication (Testable).

- evaluator plurality as embedded subsystems,
- separation of representation and evaluation,
- support for self-monitoring and refinement loops.

5.6 Prediction 6: Energetic Limits on Computation

Triadic Prediction. Computation must obey energetic constraints; systems that compensate for geometric deficiency with energy alone become inefficient.

AI Prediction. Architectures that scale compute without structural change will show diminishing returns, while architectures that balance triadic roles will maintain efficiency.

LLM Implication (Testable).

- disproportionate compute requirements for marginal gains,
- inefficient scaling at frontier sizes,
- diminishing returns from depth or width increases.

DMLM Implication (Testable).

- compute scales with modular expansion,
- efficiency preserved through structural balance,
- avoidance of diminishing returns.

5.7 Prediction 7: Preservation of Cyclic Interdependence

Triadic Prediction. Any viable system must preserve the closed interdependence

$$E \rightarrow G \rightarrow I \rightarrow E.$$

AI Prediction. Architectures that break the cycle (e.g., by lacking geometry) will exhibit structural failure modes, while architectures that preserve the cycle will scale coherently.

LLM Implication (Testable).

- geometric deficiency breaks the cycle,
- compensation through energy-like scaling,
- structural failure modes at large scale.

DMLM Implication (Testable).

- balanced triadic roles across modules,
- recursive fractal scaling,
- coherent behaviour at increasing scale.

6 Conclusion

This paper has examined the empirical limits of monolithic transformer scaling through the structural lens provided by the triadic framework. While large language models have benefited enormously from predictable scaling laws, their behaviour at frontier scales reveals consistent patterns of saturation, interference, and instability. These phenomena are often treated as engineering challenges, yet their persistence across architectures and training regimes suggests deeper structural causes.

By interpreting transformers in terms of energy-like, geometry-like, and information-like roles, we have shown that their limitations arise from a systematic imbalance: energy and information dimensions scale aggressively, while the geometric dimension remains effectively absent. This absence leads to global coupling, representational crowding, and the lack of stable

organizational structure—factors that collectively constrain the ability of monolithic models to scale coherently.

In contrast, the Dynamic Modular Language Model (DMLM) architecture illustrates how triadic balance can be restored. Through modularity, paging, evaluator plurality, and canonical forms, DMLM introduces the geometric structure required for stable long-range computation and avoids the structural bottlenecks inherent to transformers. The purpose of this comparison is not to prescribe a specific design, but to demonstrate that architectures respecting triadic balance offer a principled path beyond the limits of monolithic scaling.

The broader implication is that scaling behaviour cannot be understood solely through empirical laws or resource curves. It must be grounded in the structural organization of the underlying architecture. By applying a general triadic framework to the domain of AI scaling, this paper provides a conceptual basis for interpreting current limitations and for guiding the development of next-generation systems that scale more robustly in depth, coherence, and stability.

A Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository: <https://github.com/triadic-cosmos>.

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